

Hessian

$$f(x, y) = f(x_0, y_0)$$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

$$+ \underbrace{\frac{\partial f}{\partial x} \Big|_{(x_0, y_0)} (x - x_0) + \frac{\partial f}{\partial y} \Big|_{(x_0, y_0)} (y - y_0)}_{\nabla f \Big|_{(x_0, y_0)} \cdot \vec{h}}$$

$$+ \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial x^2} \Big|_{(x_0, y_0)} (x - x_0)^2 + \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)} (x - x_0)(y - y_0) + \frac{1}{2} \frac{\partial^2 f}{\partial y^2} \Big|_{(x_0, y_0)} (y - y_0)^2}_{\frac{1}{2} \vec{h}^t H_f \Big|_{(x_0, y_0)} \vec{h}}$$

+ ...

$$\vec{h}^t H_f \vec{h} = \begin{bmatrix} (x - x_0) & (y - y_0) \end{bmatrix} \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} \begin{bmatrix} (x - x_0) \\ (y - y_0) \end{bmatrix}$$

Hessian

The trick is to understand what do these second derivatives tell us about the function. The best place to figure out what they mean is at a critical point, that is a point where $\nabla f = 0$. For simplicity suppose that $x_0 = y_0 = 0$, and $f(x_0, y_0) = 0$, and of course $\nabla f|_{(0,0)} = 0$. Then we have that

$$f(x, y) = \frac{1}{2} \vec{h}^t H_f|_{(x_0, y_0)} \vec{h} + \cdots,$$

and so f behaves like a quadratic form near by the origin. Suppose that H_f is nondegenerate, that is, suppose that it has no zero eigenvalues. In that case it really behaves like one of three cases (after a linear change of coordinates).

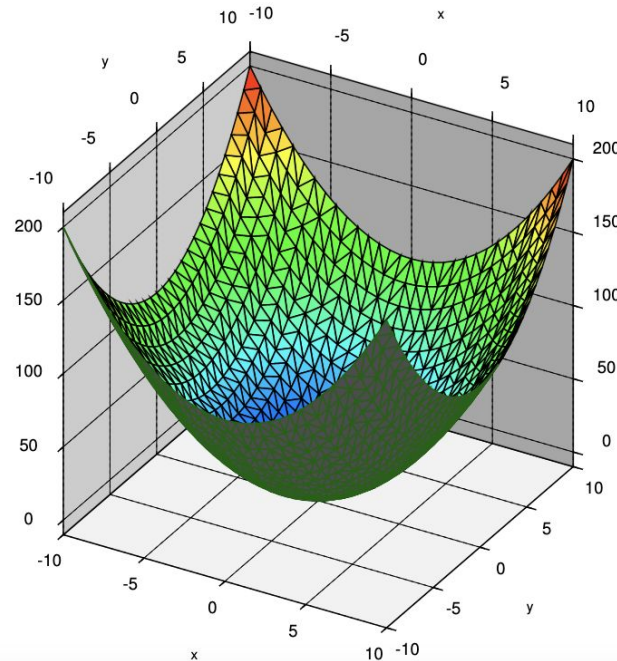
Hessian

Two positive eigenvalues:

The model case is when the matrix looks like

$$\frac{1}{2}H_f = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(we write half the matrix for simplicity). The function then becomes $x^2 + y^2$, and then the graph of the form looks like



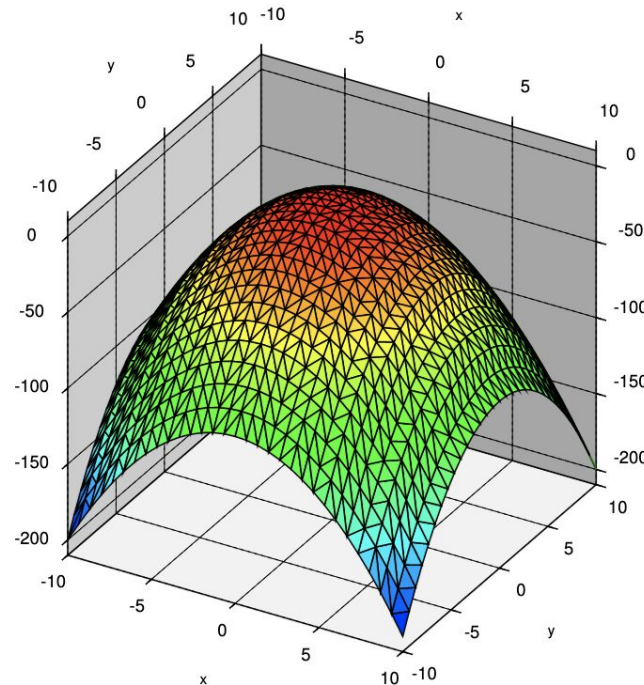
Hessian

Two negative eigenvalues:

The model case is when the matrix looks like

$$\frac{1}{2}H_f = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The function then becomes $-x^2 - y^2$, and then the graph of the form looks like



Hessian

One positive and one negative eigenvalue:

The model case is when the matrix looks like

$$\frac{1}{2}H_f = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The function then becomes $x^2 - y^2$, and then the graph of the form looks like

